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VEHICLE REAR STRUCTURE

Abstract

The vehicle rear structure includes a roof side rail having a closed cross-sectional shape, a pillar extending in a vertical direction, one end of which is connected to the rear wheel house and the other end of which is connected to a portion in front of the rear end portion of the roof side rail, and a rear framework member positioned behind the pillar and having one end connected to the rear wheel house and the other end of which is connected to the rear end portion of the roof side rail.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-031552 filed on Mar. 1, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a vehicle rear structure.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2021-116035 (JP 2021-116035 A) discloses a vehicle rear structure including a pair of right and left C pillars connecting rear portions of side sills and roof side rails.

SUMMARY

[0004] The C pillar of JP 2021-116035 A has low rigidity against a rotational moment about a rotational axis extending in a vehicle width direction. Therefore, when a vertical force applied to a rear suspension of the patent document is transmitted to the C pillar, the C pillar is likely to rotate about the rotational axis. Thus, when the vertical force is applied to the vehicle rear portion, the vehicle rear portion is likely to deform. When the amount of deformation of the vehicle rear portion is large, the steering feeling of the steering wheel for the driver and the riding comfort of the occupant decrease.

[0005] In view of the above fact, it is an object of the present disclosure to provide a vehicle rear structure that is unlikely to deform when subjected to a vertical force.

[0006] A vehicle rear structure according to an aspect includes: [0007] a roof side rail having a closed sectional shape; [0008] a pillar extending in a vertical direction and including one end connected to a rear wheel house and another end connected to a portion forward of a rear end portion of the roof side rail; and [0009] a rear framework member positioned on a vehicle rear side of the pillar and including one end connected to the rear wheel house and another end connected to the rear end portion of the roof side rail.

[0010] In the vehicle rear structure of the above aspect, the roof side rail has the

[0011] closed sectional shape. Therefore, the roof side rail is unlikely to deform in the vertical direction when subjected to a vertical force as compared with a case where the sectional shape of the roof side rail is an open section. The pillar extending in the vertical direction includes one end connected to the rear wheel house and the other end connected to the portion forward of the rear end portion of the roof side rail. The rear framework member positioned on the vehicle rear side of the pillar includes one end connected to the rear wheel house and the other end connected to the rear end portion of the roof side rail. Therefore, the pillar is unlikely to rotate about a rotational axis extending in a vehicle width direction when subjected to a vertical force as compared with a case where only the pillar is connected to the rear wheel house and the roof side rail. In this way, the roof side rail is unlikely to deform in the vertical direction and the pillar is unlikely to rotate about the rotational axis extending in the vehicle width direction when subjected to the vertical force. Thus, the vehicle rear structure of the above aspect is unlikely to deform when subjected to the vertical force.

[0012] In the vehicle rear structure of the above aspect, the rear framework member may extend in the vertical direction.

[0013] In the vehicle rear structure of the above aspect, the pillar is unlikely to rotate about the rotational axis extending in the vehicle width direction and the roof side rail is more unlikely to deform in the vertical direction when subjected to the vertical force as compared with a case where the rear framework member extends in a direction different from the vertical direction.

[0014] The vehicle rear structure of the above aspect may further include: [0015] a rear side member; [0016] a rear end connected to a rear end portion of the rear side member; and [0017] a panel positioned on the vehicle rear side of the rear framework member and connected to the rear

framework member and the rear end.

[0018] In the vehicle rear structure of the above aspect, bending of the rear side member in the vertical direction is easily suppressed by the action of the panel.

[0019] The vehicle rear structure according to the present disclosure has an excellent effect that it is unlikely to deform when subjected to the vertical force.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0021] FIG. 1 is a schematic perspective view of a vehicle showing a vehicle rear structure according to an embodiment when viewed from behind;

[0022] FIG. 2 is a schematic side view of a vehicle rear structure;

[0023] FIG. 3 is a cross-sectional view taken along line 3-3 of FIG. 2;

[0024] FIG. 4 is a cross-sectional view taken along line 4-4 of FIG. 2;

[0025] FIG. 5 is a cross-sectional view taken along line 5-5 of FIG. 2;

[0026] FIG. 6 is a cross-sectional view taken along line 6-6 of FIG. 2;

[0027] FIG. 7 is a cross-sectional view taken along line 7-7 of FIG. 2; and

[0028] FIG. 8 is a cross-sectional view taken along line 8-8 of FIG. 4.

DETAILED DESCRIPTION OF EMBODIMENTS

[0029] Hereinafter, an embodiment of a vehicle rear structure according to the present disclosure will be described with reference to the accompanying drawings. Note that an arrow FR appropriately shown in the drawings indicates the front side of the vehicle which is the front side in the vehicle front-rear direction, an arrow LH indicates the left side of the vehicle which is the left side in the vehicle left-right direction, and an arrow UP indicates the upper side of the vehicle which is the upper side in the vehicle vertical direction. In the following description, the front-rear direction, the left-right direction, and the up-down direction represent the vehicle front-rear direction, the vehicle left-right direction, and the vehicle up-down direction, respectively.

[0030] As shown in FIGS. 1 and 2, the vehicle rear structure 13 applied to the rear portion of the vehicle body 12 of the vehicle 10 includes a pair of left and right metal side sills (rockers) (not shown) which are separated from each other in the vehicle width direction (left-right direction) and extend in the front-rear direction, a pair of left and right metal rear side members 14 (only the left rear side member 14 is shown in FIG. 1) which are connected to the rear end of the side sills and extend toward the rear, and a metal cross member 16 (see FIG. 1) which is connected to the left and right rear side members 14 and extends in the left-right direction.

[0031] The vehicle rear structure 13 further includes a pair of left and right metal roof side rails 18 (only the right roof side rails 18 are shown in FIGS. 1 and 2) located above the left and right side sills and the rear side members 14 and extending in the front-rear direction apart from each other in the left-right direction, and a metal rear header 25 connecting the left and right roof side rails 18 and extending in the left-right direction.

[0032] As illustrated in FIGS. 3 to 7, the roof side rail 18 is configured by coupling the outer rail 19, the inner rail 22, and the rear side panel 29 to each other. The outer rail 19 has the same length as the entire length of the roof side rail 18. The inner rail 22 is substantially parallel to the outer rail 19 and has a shorter overall length than the outer rail 19. The outer rail 19 and the inner rail 22 are integrally formed, and are manufactured by, for example, press molding.

[0033] As shown in FIG. 3, the front portion of the roof side rail 18 includes an outer rail 19 and an inner rail 22 located inside the outer rail 19. An upper edge portion and a lower edge portion of the

outer rail **19** are fixed to the inner rail **22**.

[0034] As shown in FIGS. **4** to **6**, the portion located between the predetermined

[0035] portion and the rear end located behind the front end in the front-rear direction of the roof side rail **18** is constituted by the outer rail **19** and the upper portion of the metal rear side panel **29** located inside the outer rail **19**. The upper edge portion and the lower edge portion of the outer rail **19** are fixed to the vehicle outer side surface of the rear side panel **29**. The lower end portions of the left and right rear side panels **29** are connected to the left and right rear wheel houses **27**, respectively.

[0036] As described above, the roof side rail **18** is composed of two members over its entire length. Therefore, at any position in the longitudinal direction of the roof side rail **18**, the shape of the cross section cut in a plane orthogonal to the longitudinal direction of the roof side rail **18** is annular. That is, the roof side rail **18** has a closed cross-section over its entire length.

[0037] The front end portions of the left and right roof side rails **18** are connected to the rear end portions of the roof side rails **23** made of a metal different from the roof side rails **18**.

[0038] The vehicle rear structure **13** includes a pair of left and right metal rear wheel houses **27**. The left and right rear wheel houses **27** are connected to the left and right ends of the cross member **16**. A rear wheel (not shown) is provided in an internal space of each rear wheel house **27**.

[0039] As shown in FIGS. **1** and **2**, a metal C-pillar **30** extending in the vertical direction in a side view and positioned outside the rear side panel **29** is positioned between the left and right rear wheel houses **27** and a portion positioned forward and rearward from the rear end of the left and right roof side rails **18**. A flat connecting portion **31** is provided at an upper end portion of the C pillar **30**. The cross-sectional shape of the portion located below the connecting portion **31** of the C pillar **30** is a substantially hat shape (see FIG. **8**). A connection flange **32** is provided at a front edge portion of a portion located below the connecting portion **31** of the C pillar **30**, and a connection flange **33** is provided at a rear edge portion. As shown in FIGS. **1** and **4**, the connecting portion **31** is fixed to a portion that is in front of the rear end portion of the vehicle outer side surface of the outer rail **19** and in rear of the front end portion. The lower end portions of the connection flange **32** and the connection flange **33** are fixed to the vehicle outer surface of the rear wheel house **27**, and **20** portions except the lower end portion and the upper end portion of the connection flange **32** and the connection flange **33** are fixed to the vehicle outer surface of the rear side panel **29**.

[0040] Further, as shown in FIGS. **1** and **2**, between the rear end portions of the left and right rear wheel houses **27** and the left and right roof side rails **18**, a metal rear framework member **35** which is located at the vehicle rear side from the C pillar **30** and extends in the **25** vertical direction in a side view is located. The rear framework member **35** is located outside the rear side panel **29**. In the present embodiment, the extension direction of the rear framework member **35** is inclined with respect to the extension direction of the C pillar **30** in a side view. The cross-sectional shape of the portion excluding the upper end portion of the rear framework member **35** is a substantially hat shape. A connection flange **37** is **30** provided at a front edge portion of a portion where the cross-sectional shape of the rear framework member **35** forms a substantially hat shape, and a connection flange **38** is provided at a rear edge portion. As shown in FIG. **1**, the lower end portions of the connection flange **37** and the connection flange **38** are fixed to the vehicle outer surface of the rear wheel house **27**, and portions except the lower end portion and the upper end portion of the connection flange **37** and the connection flange **38** are fixed to the vehicle outer surface of the rear side panel **29**. Further, an upper end portion of the rear framework member **35** is fixed to the outer rail **19**.

[0041] In the present specification, the term “the C pillar **30** extends in the up-down direction” in a side view includes a case where the extending direction of the C pillar **30** is completely parallel to the up-down direction and a case where the extending direction of the C pillar **30** is substantially parallel to the up-down direction. That is, inclination of the extension direction of the C pillar **30** in a range of a predetermined angle or less with respect to the up-down direction in the side view is

included in “the C pillar **30** extends in the up-down direction in the side view”. Similarly, in the present specification, the term “the rear framework member **35** extends in the up-down direction” in a side view includes a case where the extension direction of the rear framework member **35** is completely parallel to the up-down direction and a case where the extension direction of the rear framework member **35** is substantially parallel to the up-down direction. That is, inclination of the extension direction of the rear framework member **35** in the side view within a range of a predetermined angle or less with respect to the up-down direction is included in “the rear framework member **35** extends in the up-down direction in the side view”. The predetermined angle is, for example, XX degrees.

[0042] A metal rear end **40** extending in the left-right direction is connected to the rear end portions of the left and right rear side members **14**. Further, a front end of a D pillar **42** (only the D pillar **42** on the right side in FIG. **1**) which is a metal elongated member is connected to a rear end of each of the left and right roof side rails **18**. In a side view, the D pillar **42** is curved, and a rear end portion of each D pillar **42** is connected to the rear end **40**. The opening surrounded by the rear header **25**, the left and right roof side rails **18**, the left and right D pillars **42**, and the rear end **40** is an opening that is openably closed by a back door (not shown).

[0043] Further, a pair of left and right metal shear panels **45** are provided in a space formed between the rear wheel house **27**, the rear side panel **29**, the rear end **40**, and the D pillar **42**. The rear end portion of the shear panel **45** is connected to the rear end **40**, the front portion of the lower portion of the shear panel **45** is connected to the rear wheel house **27**, and the upper portion of the shear panel **45** is connected to the rear portion of the rear side panel **29** and the front end portion of the D pillar **42**.

[0044] A rear suspension (not shown) is provided in a component (not shown) of the vehicle rear structure **13** different from the cross member **16** and the cross member **16**. A part of the components of the rear suspension is connected to the left and right rear wheel houses **27**.

Action and Effect

[0045] Next, the operation and effects of the embodiment will be described.

[0046] When a force enters the traveling vehicle **10** from the road surface, this force is transmitted from the rear suspension to the rear portion of the vehicle body **12** (the rear wheel house **27**), and the rear portion of the vehicle body **12** vibrates in the vertical direction.

[0047] However, the roof side rail **18** has a closed cross-sectional shape. Therefore, the roof side rail **18** is hardly deformed in the up-down direction when the rear portion of the vehicle body **12** is subjected to a force in the up-down direction as compared with a case where the cross-sectional shape of the roof side rail is an open cross-section.

[0048] Further, the roof side rail **18** having a long front-rear dimension (the front end being positioned forward of the C pillar **30**) receives the force transmitted from the rear wheel house **27** to the roof side rail **18** via the rear side member **14**, the rear end **40**, the shear panel **45**, and the D pillar **42** in its own axial direction (longitudinal direction). Therefore, when the rear portion of the vehicle body **12** is subjected to the vertical force, the rear end **40**, the shear panel **45**, and the D pillar **42** are hardly deformed.

[0049] Further, one end of the C pillar **30** extending in the up-down direction is connected to the rear wheel house **27**, the other end is connected to a portion of the rear end portion of the roof side rail **18** in front of and behind the front end, one end of the rear framework member **35** positioned behind the vehicle from the C pillar **30** is connected to the rear wheel house **27**, and the other end is connected to the rear end portion of the roof side rail **18**. Therefore, compared to the case where only the C pillar **30** is connected to the rear wheel house **27** and the roof side rail **18**, when the rear portion of the vehicle body **12** receives a force in the vertical direction, the C pillar **30** is hardly rotated around the rotation axis extending in the vehicle width direction, and the roof side rail **18** is hardly deformed in the vertical direction.

[0050] As described above, when the rear portion of the vehicle body **12** receives the vertical force,

the roof side rail **18**, the rear end **40**, the shear panel **45**, and the D pillar **42** are less likely to be deformed, and the C pillar **30** is less likely to rotate around the rotation axis extending in the vehicle width direction. Therefore, the vehicle rear structure **13** of the present embodiment is hardly deformed when the rear portion of the vehicle body **12** is subjected to a force in the vertical direction. Therefore, in the vehicle **10** of the present embodiment, when force enters from the road surface, the steering feeling of the steering wheel by the driver and the riding comfort of the occupant are unlikely to decrease.

[0051] Further, since the rear framework member **35** extends in the up-down direction in a side view, when the rear portion of the vehicle body **12** receives a force in the up-down direction as compared with the case where the rear framework member **35** extends in a direction different from the up-down direction in a side view, it is difficult for the C pillar **30** to rotate around the rotation axis extending in the vehicle width direction, and the roof side rail **18** is hardly deformed in the up-down direction.

[0052] The vehicle rear structure **13** further includes a rear end **40** connected to the rear end of the rear side member **14**, a rear wheel house **27**, a rear side panel **29**, and a shear panel **45** connected to the D pillar **42**. Therefore, the vehicle rear structure **13** is easier to suppress the vertical bending of the rear side member **14** than in the case where the shear panel **45** is not provided.

[0053] Further, when the C pillar **30** rotates around the rotation axis extending in the vehicle width direction, the D pillar **42**, which is separated from the C pillar **30** by a predetermined distance rearward, receives the rotation moment around the C pillar **30**. However, in the present embodiment, since the rotation of the C pillar **30** is suppressed, the rotational moment around the C pillar **30** received by the D pillar **42** is small. Therefore, when the rear portion of the vehicle body **12** is subjected to the vertical force, the D pillar **42** is hardly deformed. Further, since the D pillar **42** is hardly subjected to a large rotational moment, even if the structure of the D pillar **42** is simplified, large deformation of the D pillar **42** is hardly caused.

[0054] Although the vehicle rear structure according to the embodiment has been described above, the present disclosure can be appropriately modified in design without departing from the gist thereof.

[0055] For example, the extension direction of the rear framework member **35** in the side view may be parallel to the extension direction of the C-pillar **30**.

Claims

1. A vehicle rear structure comprising: a roof side rail having a closed sectional shape; a pillar extending in a vertical direction and including one end connected to a rear wheel house and another end connected to a portion forward of a rear end portion of the roof side rail; and a rear framework member positioned on a vehicle rear side of the pillar and including one end connected to the rear wheel house and another end connected to the rear end portion of the roof side rail.
 2. The vehicle rear structure according to claim 1, wherein the rear framework member extends in the vertical direction.
 3. The vehicle rear structure according to claim 2, further comprising: a rear side member; a rear end connected to a rear end portion of the rear side member; and a panel positioned on the vehicle rear side of the rear framework member and connected to the rear framework member and the rear end.
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